RESEARCH MONOGRAPH · REVISION 3.0

Possible Worlds, Not Panic

A Positive Theory of Irreversible Execution under Partial Observability, with Amendment 0 to the Desk Execution Protocol

Working paper in the Desk / DEP / AMS sequence · Friedman–operational frame
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Abstract

Retail execution automation fails in two characteristic ways. The first is Friedman's problem: the operator inserts administrative latency into a price process that does not wait. The second is an epistemic problem the literature usually misnames "risk management": the harness converts ignorance into a convenient story so that trading can continue. Desk v1 optimized for the comfort of the second error (halt on any desync). Desk v2 named the first error and then overclaimed cryptographic and consensus properties it did not possess. This monograph states a positive theory. An execution harness is not an edge engine. It is a machine that maintains a conservative set of venue worlds still consistent with evidence, and refuses any new side effect whose safety requires pretending one of those worlds has disappeared. The operational name of that refusal is an uncertainty budget. The testable claim is not that the book is "true," but that accounted exposure is an upper bound on every real exposure still admitted by observation. We repair three predicates in the adversarial model (WAL-before-effect, intent versus manifestation, max rather than min exposure), demote Merkle theater and longest-chain analogies, and treat NinjaTrader ATI's unobservable internal book as a priced observability tax rather than a protocol bug. Bounded model checking is evidence, not proof. The research program is to make that distinction boringly operational.

Keywords: execution microstructure · partial observability · write-ahead logging · possible-worlds exposure · positive economics of automation · NinjaTrader ATI · Webull OpenAPI

> *"The relevant question to ask about the 'assumptions' of a theory is not whether they are descriptively 'realistic,' for they never are, but whether they are sufficiently good approximations for the purpose in hand."*
>
> — *Milton Friedman, The Methodology of Positive Economics (1953)*

I. The Problem Friedman Would Name

A price is a compressed message about scarcity. It updates without asking whether the operator has clicked. Any execution design that requires a human veto on every order is therefore not a safety system. It is a tax on the information content of the quote. Friedman's intolerance for that tax is not romanticism about machines. It is a statement about opportunity cost. While the finger hovers, the allocation the price was pointing at is being taken by someone whose latency budget is smaller.

That is the first error of retail automation: *artificial latency as a substitute for a rule*. The second error is quieter and more expensive. When the harness cannot see whether an order lived or died, it prefers a story that restores permission to act. "No fill in the folder" becomes "never submitted." "Three NOT_FOUND polls" become "authoritative absence." "Local net is zero" becomes "no exposure." Each of those inferences is a theory about the world. None of them is an observation. A positive theory of execution starts by refusing to let theory

pose as observation.

Call the second error *invented knowledge*. It is the operational cousin of what Friedman attacked when economists treated untested assumptions as if they were facts because the facts were inconvenient. The harness that invents knowledge does not become safer. It becomes a machine that can double a position while reporting a clean book. The operator then discovers the position in the only place that was ever authoritative: the venue.

This paper takes both errors as empirical regularities of file-drop and HTTP retail stacks, specifically NinjaTrader 8 Advanced Trade Interface (ATI) and Webull OpenAPI. It does not argue that those venues are good. It argues that they are the environment. A theory that requires a better environment is a wish. A harness that survives this environment is a result.

II. What Desk v1 Optimized For

Desk v1 was a bureaucracy with sockets. Its revealed preference was administrative comfort. Any mismatch — wiped `outgoing/`, cross-volume `shutil.move`, HTTP 429, clock skew — became a global halt. The theory implicit in that design is: uncertainty is an emergency, and the safe action under emergency is to freeze. That theory fails two tests.

First, freeze is itself a side effect with a price. Working orders remain. Delta remains. The market continues to reprice the book the harness just refused to look at. A halt that does not flatten is not safety. It is a decision to stop updating a model while the world it models keeps moving.

Second, the halt trained the operator to treat recovery as folklore: dust the logs, rebuild state by eye, resume. That is not verification. It is a ritual that happens to coincide with being lucky. File-system folklore is especially costly on ATI, where “atomic move” across devices is a copy-delete race and a successful local rename is not a venue acknowledgment.

v1’s checklist of twenty paper sessions before live promotion is the same error in calendar form. Time is not evidence. A probe that fails on Tuesday and passes on Thursday has not become true because Thursday arrived. Friedman would have called this confusing a waiting period with a test.

III. What Desk v2 Claimed, and What It Did Not Possess

Desk v2 made the correct architectural turn. Uncertainty should be explicit, bounded, and recoverable. Session-gated authorization replaced the per-order click. Adapter states became discrete. Venue fills were named as more authoritative than local hope. That turn is kept.

The prose then spent the surplus on metaphors it could not cash. A hash-linked journal printed to stdout is local tamper evidence. It is not an anchor. Software that can rewrite the journal can rewrite the printed root. A venue fill tape is an external execution authority. It is not a longest chain. Nakamoto consensus is a specific mechanism with a fork-choice rule and an incentive system. A broker is a firm with a database and a lawyer. Borrowing the word “consensus” does not purchase the mechanism.

Three further overclaims matter because they survive into code if uncorrected. A bounded in-memory LRU of client order IDs is not restart-safe exactly-once semantics. Continuing to trade on local state during a partition is how duplicate economic intent is born. Treating missing covariance as zero correlation is not conservative; it is a silent narrowing of the possible world set. Kelly-like sizing inside the kernel is a category error: the harness

claims it does not possess edge, then sizes as if it does.

The useful remainder of v2 is a division of labor. The kernel enforces hard envelopes. Strategy may, as a plugin, resize inside those envelopes. BOOK is authoritative for what Desk believed, decided, attempted, observed, and reconciled. The venue is authoritative for what the venue will still honor. Positions and cash are derived projections. Never rewrite BOOK to agree with a venue. Append the disagreement.

IV. A Positive Theory

The unit of analysis

The unit is not "an order." The unit is a lineage of identities that must not be collapsed:

```
EconomicIntent → ExecutionIntent → DispatchAttempt(s) → VenueManifestation(s) → Fill(s)
```

An economic intent is the signed claim about desired exposure. An execution intent is that claim after risk has quantized it and after it has been durably committed. A dispatch attempt is one try to make the outside world hear it. A venue manifestation is whatever object the venue created, if it created one. A fill is a consequence. Identity is not attempt. Attempt is not manifestation. Manifestation is not consequence. HTTP APIs that honor a durable `clientOrderId` may collapse many attempts into one manifestation. ATI file drops may not. The model must not pretend they are the same transport.

The epistemic primitive

Let O be the set of observations Desk currently holds: journaled commits, acknowledgments, query snapshots, explicit rejects, operator attestations. Let $\Omega(O)$ be the set of venue worlds still consistent with O. Desk does not need to know which world is actual. It needs to refuse actions that are safe in only a proper subset of $\Omega(O)$.

Constitutional theorem (possible-worlds conservatism).

> MAG/DEP may narrow $\Omega(O)$ only when evidence justifies the narrowing. Resource governance must remain safe across every world still admitted by the system's evidence. Equivalently: accounted exposure A on instrument i must satisfy $A_i \geq \sup_{\omega \in \Omega(O)} R_i(\omega)$, where $R_i(\omega)$ is real signed (or gross, as policy requires) exposure in world ω. The useful invariant is an upper bound on the worst remaining world, not a lower bound on the kindest one.

A signed scalar is not a sufficient statistic. Uncertain BUY 100 and uncertain SELL 100 can net to zero while Ω still contains +100 and −100. The state the kernel must carry is an interval, or a set:

```
E_i^{possible} = [E_i^{min}, E_i^{max}] worst_long = max(E_possible) worst_short =
|min(E_possible)| gross_risk = max(worst_long, worst_short)
```

Portfolio constraints are harder because uncertain executions are jointly distributed. That is not a reason to return to a scalar. It is a reason to build a simulator whose state is the set, not a reason to pretend the set is a point.

Uncertainty as a priced resource

"Uncertainty consumes risk capacity" is the slogan form. The precise form is: every world still in Ω reserves the exposure that world would realize. Liveness is subordinate to that reservation. An adapter that cannot shrink Ω — ATI after an `outgoing/` wipe, a venue that will not answer queries — is allowed to exhaust its uncertainty budget and refuse new intents. That is not a protocol failure. It is the price of insufficient observability, stated in the only unit the operator already understands: size he is no longer permitted to add.

This is a Friedman move. A constraint that cannot be seen will be paid somewhere. Making the tax visible is the beginning of a market in observability. If ATI's write-only character costs too much uncertainty budget, the operator switches venues or accepts human attestation as the only authoritative-negative evidence that adapter can produce. The protocol does not invent a third option.

V. Amendment 0 to DEP-1 / AMS

DEP-1 was the right next artifact and is not implementable as written. Amendment 0 changes the constitution, not the feature list.

P9 repaired

Accounted exposure is a conservative upper bound on possible real exposure. The written predicate that compared against a minimum possible outcome is struck. Checkers compare A to the supremum of remaining worlds. Interval exposure replaces scalar netting as the default representation.

P1 repaired: durability is a clock, not a log membership test

"Intent exists in the journal" does not prove the intent existed before the side effect. A harness can POST, then journal STAGED, then journal DISPATCH, and still look well-formed after the fact. The model introduces explicit observation points around every durability boundary:

```
CREATE_INTENT → APPEND_STAGE → FSYNC_STAGE → SIDE_EFFECT → APPEND_DISPATCH → FSYNC_DISPATCH
```

The invariant is temporal. At the instant any external side effect becomes possible, the corresponding ExecutionIntent already exists in stable storage. The model distinguishes process memory, OS page cache, and durable media. `append()` is not `fsync()`. A crash between them is a first-class adversary action.

P4 repaired: attempts may duplicate; manifestations may not hide

Two DispatchAttempts sharing an EconomicIntent identifier are permitted. Two unaccounted VenueManifestations of the same economic intent are not. Idempotent HTTP with a durable client identifier may collapse attempts. ATI may produce two files, two working orders, or one file the platform never ack'd. Recovery never resubmits an IntentID whose manifestation set is still open. A new economic intent, if the operator wants another try, consumes additional budget until the first lineage is closed by evidence.

Evidence strength replaces poll numerology

Three identical NOT_FOUND replies increase confidence. They do not mint authoritative negative evidence unless the venue contract says they do. DEP transitions are indexed by evidence class:

Class	Meaning	May shrink Ω?
NON_AUTHORITATIVE	Local inference, timeout, empty folder, stale UI	No
EVENTUALLY_CONSISTENT	Repeated compatible queries, no contract of finality	Only toward caution, never toward release
AUTHORITATIVE_POSITIVE	Venue reports a live order, fill, or working qty	Yes: add that manifestation to the known set
AUTHORITATIVE_NEGATIVE	Venue contractually asserts absence, or operator attestation on a write-only adapter	Yes: drop worlds that required the missing object

Table 1. Evidence classes. Release of reserved capacity requires authoritative negative or an authoritative positive that fully explains the reservation.

Adversary completion

Loss, delay, stale snapshots, and process death are necessary and not sufficient. Amendment 0 adds duplicate submission, duplicate acknowledgment, duplicate fill, out-of-order events, contradictory query results, and late events after a locally terminal belief. Two traces are mandatory in any conformance suite:

```
T1 PLACE → TIMEOUT → CANCEL → CANCEL_ACK → LATE_FILL(parent) → RECONCILE T2 REPLACE → parent
fills → replacement fills → net exposure visible
```

DEP already admits both races. AMS must prove the resulting interval is never quietly netted to a false zero.

What a passing run means

A twenty-step exhaustive exploration plus ten thousand randomized trajectories is a serious test. It is not a proof of the protocol. The sentence “if AMS passes, we have proven…” is struck. A passing campaign establishes that no counterexample was found inside the modeled failure semantics and the explored bounds. The artifact is a conformance report, not a trophy runtime:

protocol and model version hashes; random seeds; explored state count; transition count; maximum depth; state and transition coverage; invariant evaluation counts; unreachable states; adapter capability matrix; counterexamples and their minimized traces; explicit assumptions; unexplored or unbounded dimensions.

VI. Operational Protocol after Amendment 0

Commit-before-effect

No external submission may occur unless its ExecutionIntent has first been durably committed. That is write-ahead logging for orders. On restart the router reconstructs open lineages from BOOK. It does not consult an LRU as source of truth. A persisted LRU, if it exists at all, is an acceleration hint and is discarded when it disagrees with the journal.

State machine, with UNCERTAIN load-bearing

DRAFT becomes STAGED only after the exposure interval of the new intent, added to every still-possible world, remains inside the envelopes. STAGED becomes IN_FLIGHT at the first moment a side effect is possible. IN_FLIGHT becomes LIVE only on authoritative-positive receipt, UNCERTAIN on timeout or crash-shaped silence, REJECTED_TERMINAL only on authoritative rejection. Exit from UNCERTAIN is a query, an attestation, or the passage of time coupled to an authoritative-negative contract. It is never the assumption that silence means absence.

Reconciliation

Reconciliation is observer correction, not book-cooking. New submissions on the affected venue pause. Snapshots are taken with whatever query surface the adapter actually has. Diffs append RECONCILE events. Local state is a projection. If a snapshot is empty and the adapter is write-only, the lineage stays UNCERTAIN and keeps its reservation. DEP-1’s clause that “not found + already UNCERTAIN ⇒ REJECTED_TERMINAL” is struck for ATI unless a REQUEST command or a human attestation supplies authoritative negative evidence.

Kernel envelopes (no Kelly inside)

The kernel enforces, and only enforces: max gross, max net, max per symbol, max order size, max daily realized loss, max daily total loss, and max outstanding unknown exposure — the last computed from the possible-world interval, not from a signed net. Dynamic sizing is a strategy plugin. If the plugin is absent, size is the operator's static cap. If covariance is stale, correlation is not zero. It is a configured worst-case or the instruments are treated as perfectly correlated for the purpose of the gross ceiling.

Venue capability matrix

Capability	Webull OpenAPI (sandbox/live)	NT8 ATI
Durable client ID	Yes, if honored	Only if encoded in PLACE text; platform need not key on it
Authoritative query of working orders	GET /orders by client id	REQUEST if enabled; else outgoing/ folklore
Authoritative negative	Only if contractually final; else eventual	Generally no; human attestation or REQUEST
Ack ≠ fill	Preview / accept distinct from execution	Rename success ≠ processed
Duplicate manifestation risk	Lower if client ID is unique and honored	Material: two files, lost outgoing, internal book
Uncertainty tax	Bounded by query RTT and consistency lag	Can consume the entire envelope

Table 2. Adapter capabilities are part of the theory, not footnotes.

VII. Testable Implications (Positive Economics)

A theory earns its keep by implications that could be false. The following are stated so they can fail.

I1. For every crash between FSYNC_STAGE and SIDE_EFFECT, replay produces no venue manifestation whose ExecutionIntent is absent from durable storage. Counterexample: any POST-then-journal ordering that AMS cannot detect.

I2. For every open lineage, accounted interval exposure dominates real exposure in every world still consistent with O. Counterexample: uncertain buy and uncertain sell netted to zero while a one-sided fill remains possible.

I3. Duplicate or late fills after a local CANCEL_ACK never disappear from the interval until an authoritative event closes them. Counterexample: trace T1 leaving reserved capacity equal to zero.

I4. An ATI adapter without REQUEST and without operator attestation never auto-releases IN_FLIGHT or UNCERTAIN reservations on the basis of an empty `outgoing/`. Counterexample: DEP-1 as first written.

I5. Two processes cannot both hold the incoming-directory lock. The loser refuses to start. Counterexample: two Desks, two PLACE files, one account.

I6. Stale covariance never reduces measured portfolio concentration relative to the last good matrix or the configured worst case. Counterexample: the v2 "treat as uncorrelated, cap at 50%" rule when the names are the same factor.

These are not moral claims. They are predictions about what a correctly instrumented harness will and will not do under a named adversary. If AMS or production telemetry falsifies one of them, the theory shrinks. That is the point of writing them down.

VIII. What This Is For (MAG), and What It Is Not

Desk is a hostile laboratory for a more general side-effect protocol. The abstraction that survived contact with the reviews is not Intent → Attempt → Observation → Belief. It is:

```
INTENT → COMMITMENT → ATTEMPT → MANIFESTATION → CONSEQUENCE → OBSERVATION → BELIEF →
(possible-world set) → UNCERTAINTY BUDGET → RECONCILIATION
```

Any system that spends irreversible action through an unreliable actor — deploying software, wiring money, sending a legal notice, moving a robot arm — has this shape whether it admits it or not. Trading is merely the domain in which a wrong story about the world is priced in minutes rather than quarters. That is why it is a good lab. It is not why the lab is a trading strategy.

The harness does not generate edge. It will execute a bad hypothesis flawlessly into a drawdown and then produce a journal that shows exactly which rule did it. That is the product. Operators who want a shield against reality should not buy a clock.

IX. Limits, Stated so They Cannot Hide

ATI can place internally and lose `outgoing/`. No local protocol detects that manifestation. The honest representation is write-only plus uncertainty tax, or REQUEST, or a human looking at the NT8 UI and signing an attestation. There is no fourth thing.

Webull snapshots can be stale. Repeated compatibility is not finality. Live promotion still requires the operator's separate signature (`arm.live.sig`) because probes test the machine they can see, not the lawyer they cannot.

Merkle roots without an external pin are a checksum for a careful operator, not a commitment scheme against the software itself. If that stronger claim is wanted, roots leave the trust domain — an append-only remote, a signed weekly export the operator actually verifies on a second machine, a paper note. Printing to stdout is not that.

Bounded search is not a general proof. Joint uncertainty across many names explodes. The research program is to keep the state representation honest as the set grows, not to announce completeness at ten thousand random walks.

X. Research Program

Do not write venue adapters next. Do not write the full simulator against the unamended predicates. Write Amendment 0 as the constitution — this paper is that constitution in Friedman's dialect — then a small AMS whose only job is to search for counterexamples to I1–I6. When a counterexample appears, minimize the trace, change the protocol or the claim, and only then grow the model.

After the constitution is executable, the build order is the one v2 already had and should keep: BOOK kernel with durable commit points; atomic file coordinator that refuses EXDEV; StateManager that throws into recovery instead of `sys.exit`; a localhost dashboard that shows the possible-world interval, not a single P&L number painted green; T2 flatten in the UI; T3 off the UI. The Network Shredder test is restated: for every interruption between journal commit, external submission, acknowledgment, fill, cancellation, and reconciliation, replay produces no unaccounted exposure and no duplicate economic intent.

That criterion is harder than “it comes back up.” It is also the only one that matches the theory. A harness that recovers into a lie has not recovered.

XI. Close

Friedman asked whether assumptions were good enough for the purpose in hand. The purpose in hand is not to feel certain. It is to stop spending size on the basis of a world the evidence no longer uniquely selects. Invented knowledge is cheap to write and expensive to clear. Possible-world conservatism is expensive to compute and cheap to survive.

Desk v2 pointed at the right enemy — halt-as-theology — and then dressed the pointing in consensus language the mechanism did not earn. DEP-1 made uncertainty a first-class state and then let a few convenient inferences back in through the side door. Amendment 0 nails the door. The market will not notice. The book will.

End of monograph. Constitution before adapters. Evidence before release. Worst remaining world before net.

Selected references and working corpus

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Desk working corpus (this sequence): Desk v2 specification; review of v2 truth semantics; DEP-1; review of AMS predicates P1/P4/P9; this Amendment 0 monograph.